Project Report

Campus Equipment Loan System

Xuerui Huang, Bin Wang and Yonghui Dai

Release updated 24 October 2026. This report combines a working reference product with an explicitly simulated classroom case. The current GUI, CLI, database changes and assistant-run technical checks are actual artifacts. Meeting and sprint examples do not establish student participation, client agreement or Kaiārahi endorsement.

1. Project purpose and current scope

The product supports a campus equipment desk that checks availability, records borrowing and registers returns. The current inventory contains Laptop, Camera, Projector, Tablet and Microphone. Each new loan requires a borrower name/alias and phone contact; history and contact details remain available after a restart.

A Tkinter desktop interface and the original terminal interface share LoanService and a local SQLite database. This is a standalone, single-workstation application. There is no remote server, automatic cross-computer synchronisation, login, payment or reservation workflow. Demonstrations use fictional details.

2. Team roles and proposed responsibilities

Member	Assigned role in the case	Presentation
Xuerui Huang	Requirements, coordination, menu interface, priorities, cost and report integration.	Scope, requirements, teamwork and cost.
Bin Wang	Domain objects, borrowing/return rules, integration and demonstration.	Object-oriented design and product demo.
Yonghui Dai	SQLite persistence, tests, evidence, consultation preparation and reflection coordination.	Testing, cultural considerations and reflection.

Client role and evidence boundary

The Equipment Desk Operator is a written client persona. Bin may read that role while Xuerui gathers needs and Yonghui records decisions, then return to the developer role. It is an internal rehearsal arrangement. The project owner requested the GUI, Windows support, Tablet, Microphone and phone field on 24 October; those requests are distinct from the fictional meeting records.

Timeline and effort model

The baseline case uses Day 1 kickoff, Day 2 requirements, Sprint 1 on Days 2-5 and Sprint 2 on Days 6-9, with Day 10 packaging/rehearsal. The baseline estimate is 54 hours. Section 7.1 describes later technical changes; the current-scope planning estimate adds 12 hours, totalling 66 hours. None of these estimated student hours is a timesheet.

3. Current requirements and acceptance criteria

These six stories describe the current release. Original case priorities and role assignments remain proposals for team review. Current technical results appear in section 11; acceptance by an actual client or adviser is separate.

US01 | View equipment | Must

As an operator, I want to see equipment availability so that I can respond to a loan request.

Acceptance: Laptop, Camera, Projector, Tablet and Microphone each show a unique ID, name and correct Available or On loan status.

US02 | Record borrowing and contact | Must

As an operator, I want to record a loan so that equipment use is tracked.

Acceptance: A valid device ID, alias and required phone contact create one active loan. Reject unavailable/unknown devices, invalid aliases and missing/invalid phones without adding a record. Phone input contains 7-15 digits; spaces, hyphens, parentheses and one leading + are allowed (maximum 30 characters).

US03 | Register a return | Must

As an operator, I want to register a return so that the device can be lent again.

Acceptance: Returning an active loan records a timestamp and restores availability. Reject unknown or already returned loans; keep the history.

US04 | Review loan history | Should

As an operator, I want loan history so that I can check previous and current use.

Acceptance: Show loan ID, device, alias, phone, borrowing time and active/returned status. Old records without a phone show Not recorded. Explain when no records exist.

US05 | Preserve records and upgrade data | Must

As an operator, I want saved records to survive a restart so that loan information is not lost.

Acceptance: Reopening the same database preserves history and availability without duplicating the five seed devices. Automatically upgrade the earlier schema without deleting loans or inventing phone contacts.

US06 | Preserve Unicode aliases | Must

As a user with macrons or other Unicode characters in my alias, I want them preserved so that my alias is represented accurately.

Acceptance: Save and retrieve a fictional alias such as Māia unchanged, including after restart. Do not request ethnicity; collect only the phone contact required by US02 alongside the alias.

Phone validation is a format check: 7-15 digits, at most 30 formatted characters, allowing spaces, hyphens, parentheses and one leading +. It does not verify ownership or whether the number exists. Direct community impact has not been established for this undeployed demonstration; obtain affected-user input before real operation.

3.1 Product backlog and priorities

The complete story text and acceptance criteria are in section 3. This table consolidates the current editable product backlog. The baseline sprint column identifies the original plan; GUI, extra inventory and phone changes are the later extension described in section 7.1.

Story	Priority / value	Scenario lead	Baseline sprint	Current technical checks
US01	Must: enables loan decisions	Xuerui Huang + Yonghui Dai	1	AT01, AT11
US02	Must: core workflow; prevents conflicts	Bin Wang + Xuerui Huang	1	AT02-AT04, AT12
US03	Must: completes the loan lifecycle	Bin Wang	2	AT05-AT06, AT11
US04	Should: supports accountability	Xuerui Huang + Yonghui Dai	2	AT07, AT12
US05	Must: prevents data loss	Yonghui Dai	1-2	AT08, AT12; migration unit check
US06	Must: respectful text handling	Xuerui Huang + Yonghui Dai	1-2	AT09

Ordering and scope decisions

Availability and borrowing support the initial decision; returning completes the cycle. Persistence, phone validation and alias preservation are required baseline behaviours of the current product. History is a Should item for accountability and is implemented. Client priority confirmation and cultural review remain open.

Recorded later requests

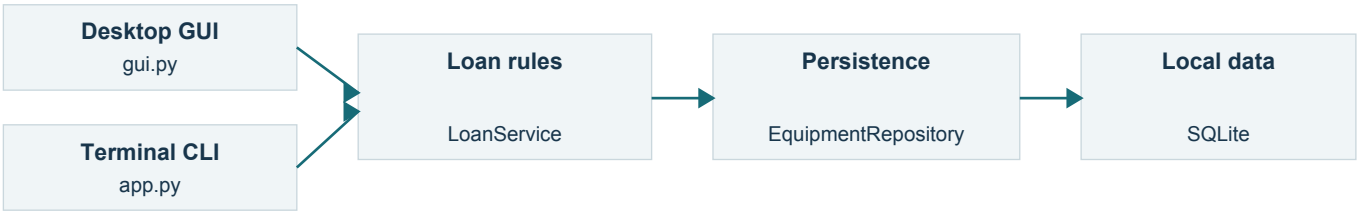
The project owner requested a GUI and Windows support, then confirmed Tablet and Microphone and a required phone contact. These changes are implemented and tested locally. They are not attributed to the earlier fictional client meetings or to an unrecorded student sprint.

Deferred CF04: online booking

Booking remains outside scope. It would require date/conflict rules and access decisions. A web version with shared records would also need a server and deployment plan; this release remains a desktop product.

Editable agile/product_backlog.md and agile/sprint_backlog.md are included in the product ZIP. They contain the same current requirements, baseline tasks and revised estimate as this report.

4. Object-oriented design and specification



Equipment and Loan objects carry data, including the borrower phone contact.

LoanApp displays the desktop interface; app.py retains the terminal workflow. Both delegate to LoanService, which receives an EquipmentRepository object. Equipment and Loan are immutable data classes. Composition keeps SQL outside the interface and validation separate from presentation.

Component	Current responsibility
gui.py / LoanApp	Equipment and history tables, name/phone input, borrow/return buttons, refresh, active filter and compact-window layout.
app.py	Equivalent terminal prompts and loan-history output, including phone contacts.
models.py	Equipment carries ID, name and availability; Loan carries record ID, equipment name, borrower, timestamps and borrower_phone.
service.py	Validate aliases and phone format; reject unknown/on-loan equipment; create loans and return active loans.
repository.py	Seed five devices, preserve SQLite records, upgrade older schemas and enforce one active loan per device.

Data model and migration

equipment(id, name) is linked to loans(id, equipment_id, borrower, borrower_phone, borrowed_at, returned_at). A null returned_at marks an active loan. A partial unique index protects the one-active-loan rule, and SQL parameters carry entered values. Timestamps are stored in UTC. Availability is derived from active loans.

On opening an older database, the repository checks the loan columns and adds borrower_phone with an empty default if absent. Earlier rows, identifiers and timestamps remain unchanged; empty historical contacts display as Not recorded. Tablet and Microphone use fixed IDs 4 and 5 and are inserted without duplicating existing seeds.

Runtime

Python 3.10+ with sqlite3; GUI operation also requires Tkinter and a desktop display. There are no third-party Python dependencies. Default storage is equipment.db beside the source, shared by GUI and CLI; --db selects another writable path. Mac and Windows launchers are included.

5. Baseline kickoff and requirements records

SIMULATED CASE RECORD. M01-M02 and decisions D01-D04 belong to the earlier written example. They are not meetings attended by the named students. Later changes are recorded separately in section 7.1.

M01 | Day 1 | Kickoff | 25-minute example

Xuerui facilitates, Bin reads the fictional operator role and Yonghui records actions. The agenda covers the equipment workflow, users and data, priorities, Te Tiriti considerations, acceptance and next steps. Each role has an opportunity to question assumptions.

Baseline decision	Reason / status in the example	Case owner
D01: limited scope	Originally a terminal and three devices. The current product later adds a GUI and two devices; see section 7.1.	Xuerui
D02: demo data	Originally alias/timestamps only. The later phone requirement supersedes the no-contact baseline; fictional contacts are used in demonstrations.	Yonghui
D03: quality	Reject double-lending and preserve saved records across restarts.	Bin
D04: cultural review	Prepare an appropriate advisory route before treating cultural criteria as settled. Review remains open.	All

M02 | Day 2 | Requirements | 20-minute example

Question in the example	Fictional operator response
How should a return identify a record?	Use the Loan ID. Equipment IDs identify devices; loan IDs identify borrowing events.
Should a returned loan disappear?	Keep the history and timestamps. The current release also retains the recorded phone.
What matters first?	Availability, borrowing, return and persistence; preserve entered aliases.
Was a web interface needed?	A terminal was sufficient for the baseline demonstration. The later desktop extension improves presentation without adding a server.
How will it be checked?	Link each story to tests; demonstrate the workflow and keep cultural review separate.

Actions and culturally informed practice

The example assigns Xuerui to consolidate needs, Bin to review rules and Yonghui to connect criteria with checks. Shared agendas, turn-taking and written follow-up support participation. The facilitator restates the ID distinction. These are case practices, not evidence of student communication or adviser consultation.

6. Sprint 1 and first review

SIMULATED CASE RECORD. Relative days, discussions and role outcomes below form a written classroom example. They are not records of meetings attended by the named students.

Baseline case: the 54-hour plan predates the GUI, new devices and phone. Sections 7.1 and 10 cover the later technical changes and revised estimate.

Goal, Days 2-5: review inventory, borrowing, storage and alias handling. This effort allocation also includes the Day 1 kickoff. The sequence illustrates development and review of the reference product; it is not reconstructed student history.

Task / story	Owner / hours	Scenario result
S1-01: Kickoff and requirements	Xuerui / 5	Scope, six stories and questions recorded.
S1-02: Inventory/borrow menu; US01-US02	Xuerui / 3	Equipment selection and success/error messages reviewed.
S1-03: Objects and borrow rules; US02	Bin / 6	Models, alias validation and duplicate protection reviewed.
S1-04: Cross-component peer review	Bin / 1	Menu/service/storage boundaries checked.
S1-05: Storage foundation; US05	Yonghui / 5	Schema, fixed seeds and active-loan index reviewed.
S1-06: Initial checks; US01-US02-US06	Yonghui / 2	Inventory, invalid/duplicate requests and macrons mapped to tests.
S1-07: Cultural preparation	Yonghui / 1	Consultation brief prepared; no adviser acceptance claimed.

M03 | Day 5 | Client-role review | 20-minute example

The review demonstrates available inventory, a successful loan and rejection of a second active loan. The operator role asks how to identify the correct loan at return and whether previous borrowing records remain visible.

Feedback	Case decision	Requirement / check
CF01: avoid double-lending	Retain both service validation and the database index.	US02 / AT03
CF02: identify the right loan	Retain the displayed Loan ID and explicit return prompt.	US02-US03 / AT02, AT05
CF03: keep earlier records	Preserve return timestamps and show record status.	US03-US04 / AT05, AT07

Retrospective and action

The example exposes a communication risk: “equipment ID” and “loan ID” can be used interchangeably in discussion. Xuerui standardises those terms in the demonstration; Bin checks the operation from menu to service; Yonghui adds the explicit return step to the review. This is a scenario improvement to explanation, not a claim that a previously observed software defect was repaired.

Effort model: Xuerui 8 h + Bin 7 h + Yonghui 8 h = 23 person-hours. The review outcome is recorded only within the simulation.

Scenario board: To do at planning; In progress during the iteration; Review at M03; complete within the written example. Cultural review remains open. No dated student board is represented.

7. Sprint 2 and final client-role review

SIMULATED CASE RECORD. Relative days, discussions and role outcomes below form a written classroom example. They are not records of meetings attended by the named students.

Baseline case: the 54-hour plan predates the GUI, new devices and phone. Sections 7.1 and 10 cover the later technical changes and revised estimate.

Goal, Days 6-9: complete the return/history explanation, review persistence and input handling, and prepare acceptance. This effort allocation also includes final packaging and rehearsal on Day 10.

Task / story	Owner / hours	Scenario result
S2-01: Return/history menu; US03-US04	Xuerui / 2	Loan ID prompt and record status reviewed.
S2-02: Cost and report integration	Xuerui / 4	54-hour plan and story/decision links reconciled.
S2-03: Workflow checks and rehearsal	Xuerui / 4	Operator steps and explanations rehearsed in the case.
S2-04: Return rules; US03	Bin / 2	Unknown/repeated returns rejected; history retained.
S2-05: Integration, design and review	Bin / 9	Integration 3 h; design explanation 3 h; review/rehearsal 3 h.
S2-06: Acceptance checks	Yonghui / 3	Story-to-test mapping and acceptance status reviewed.
S2-07: Evidence, culture and reflection	Yonghui / 7	Evidence/rehearsal 3 h; reflection/presentation 4 h.

M04 | Day 9 | Final review | 25-minute example

The operator role observes borrow, duplicate rejection, return, history and restart. Within the scenario, this covers the agreed local workflow. The role then asks for online booking. The team defers it because booking introduces dates, conflicts and further access decisions beyond the baseline.

Acceptance item	Worked-example disposition
US01-US06	Scenario workflow accepted by the fictional operator role. Actual technical results are recorded separately in section 11.
CF04: online booking	Deferred to a later iteration; not implemented or included in the NZD 1,815 baseline.
Cultural / operational acceptance	Open. No adviser review, community acceptance, real-data approval or deployment sign-off is represented.

Retrospective and next iteration

The case checks the earlier action by returning the displayed loan rather than entering the equipment ID. It still depends on one computer and demonstrator. Rehearse the extracted ZIP on the presentation computer and have each member explain a component owned by another role.

Effort model: Xuerui 10 h + Bin 11 h + Yonghui 10 h = 31 person-hours. Combined total: 54 h, with 18 h per role. These are estimates, not measured student effort.

Scenario board: To do at planning; In progress during the iteration; Review at M04; complete within the written example. Cultural review remains open. No dated student board is represented.

7.1 Current technical extension

Actual project-owner requests, 24 October 2026. The implementation and checks were performed with assistant support. These additions follow the baseline case; they are not a reconstructed third student sprint.

Change	Implemented behaviour / verification
GUI and platform launchers	Tkinter equipment/history tables and actions; Start_GUI.command for Mac and Start_GUI.bat for Windows. GUI execution and layout were checked on macOS; Windows execution is not yet verified.
Two more devices	Tablet (4) and Microphone (5), taking the fixed inventory to five. AT11 checks borrowing and returning both.
Required phone contact	GUI/CLI inputs, service format validation, saved phone column and displayed history. AT12 checks invalid input and restart persistence; unit checks cover legacy-schema preservation.

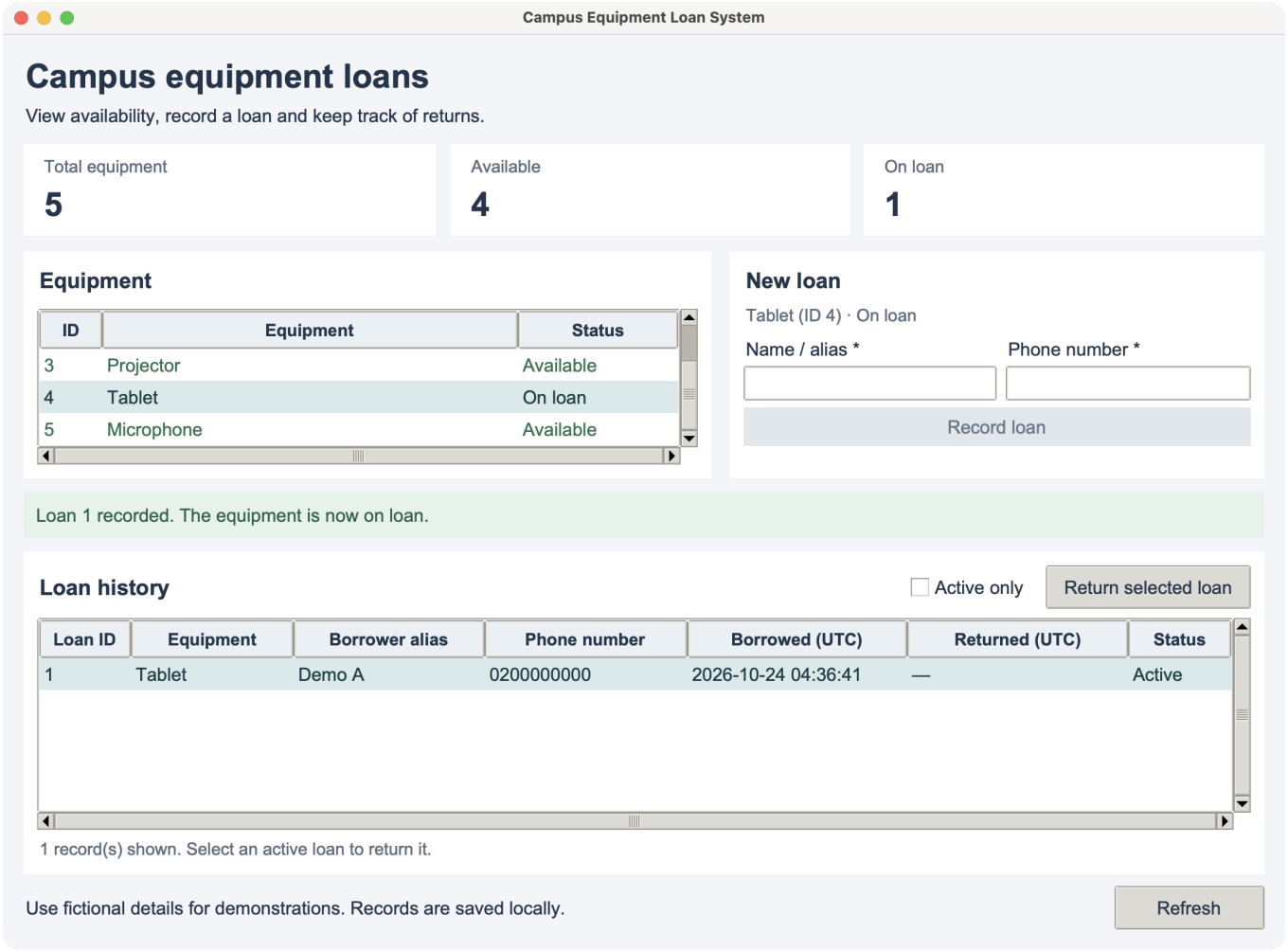


Figure 1. Actual macOS GUI demonstration using an in-memory database and fictional contact details. The equipment list scrolls; the screenshot shows the new devices. No existing borrower data is displayed.

Planning allowance: 12 additional hours, suggested as 4 per member for adaptation/review and evidence, bringing the model to 66 hours. It is an estimate, not a measured record of student work.

8. Kaiārahi consultation preparation

PREPARED; CONSULTATION OPEN. No Kaiārahi or Māori representative supplied this material. The briefing and proposed gates are not attributed advice or endorsement.

Proposed arrangement

Yonghui coordinates a request through the tutor; Xuerui explains current requirements; Bin demonstrates GUI borrowing and return with fictional details. Explain the newly required phone field and ask whether that collection and workflow are appropriate. Agree how guidance may be recorded and whether follow-up or compensation arrangements apply.

Question for an appropriate adviser	Current position to review
Who should participate?	Identify affected users and a suitable advisory route. The internal persona cannot represent Māori perspectives.
Are names and wording suitable?	Preserve Unicode aliases and macrons; do not add unverified translations or cultural imagery.
Is a phone contact necessary?	It is a current owner requirement. Review purpose, alternatives and the minimum information needed before real operation.
How should contacts be protected?	The prototype stores contacts locally without login or encryption. Agree access, retention and deletion controls before using real details.
Could the GUI exclude people?	Review keyboard access, text, window size and completion barriers; allow spoken or written feedback.
What would acceptance require?	Agree observable criteria and record any change to the product or contact requirement.

Te Tiriti principles

Partnership means seeking appropriate input before treating cultural decisions as settled. Participation means an opportunity to influence requirements and review the interface. Protection includes preserving names, minimising demonstration data and examining the new contact-data risk. These are proposals under the assessment framework, not claims of completed partnership.

Open gate	Evidence needed before closure
CG01: names and wording	Appropriate wording review; AT09 supports only character preservation.
CG02: contact data	Agreed purpose, necessary fields and controls for any real phone numbers.
CG03: participation/usability	Relevant user feedback, barriers, agreed responses and a demonstration on the presentation computer.

Record actual contact dates, adviser role, permitted summary, resulting decisions and follow-up only after consultation. Windows execution and operational acceptance also remain open.

9. Collaboration and release source record

The release source commit below contains the 13 product files. They were compared byte for byte with the staged product. Reports, presentation script and the ZIP are published in a subsequent documentation commit, avoiding a circular claim that a report contains its own commit identifier.

Field	Verified value
Repository	https://github.com/dreamgithub0920/Campus_Equipment_Loan_System
Account / branch	dreamgithub0920 / main
Source commit	d61c66ca11d2fd69c2bd40c005d3e4b53f65b100
Author	dukedyh <dukedyh@gmail.com>
Committer	dukedyh <dukedyh@gmail.com>
Author time (UTC)	2026-10-24T07:33:45Z
Author time (Pacific/Auckland)	2026-10-24T20:33:45+13:00
Message	Add desktop GUI, Windows launcher, five-device inventory and borrower phone contacts
Product path / visibility	Repository root / public

The preserved history includes 689ecdfa50594ca973a0eabafbcd04ae3ba970a7 (original terminal upload) and 40b491d129cac70bfc8e96ef449be01132e172bb (README update), both dated 21 October 2026. The new source commit is their descendant. Earlier authors, dates and parent links were not rewritten.

Communication and review model

The classroom case proposes Teams for agendas/messages, Markdown backlogs for allocation and Git for versions. Xuerui shares questions and decisions, Yonghui records actions and Bin connects behaviour to stories. Written follow-up remains available. Actual student meetings and peer review still need participant evidence.

Definition of Done and release scope

A real team task reaches Done when implemented behaviour passes relevant checks, another member reviews it and evidence is linked. The current assistant-run checks establish technical observations only. Client sign-off, Windows execution and cultural acceptance remain separate statuses.

Runtime databases, cached bytecode and .DS_Store are excluded from the uploaded release. The repository ignore rules prevent these from being added routinely. Appendix B supplies source retrieval and platform-specific start commands.

10. Current cost estimate and delivery risks

Planning assumptions: NZD 30 per hour, existing equipment and communication-tool access, and no additional paid software. This is a coursework estimate, not a rate quotation, payment record or measured student timesheet.

The baseline two-sprint model is 54 hours (18 per member). Add a 12-hour allowance for adapting/reviewing the GUI, platform launchers, new devices, phone collection, migration and evidence. A suggested 4 hours per member produces 22 hours each and 66 hours overall.

Item	Calculation / assumption	NZD
Xuerui Huang: labour	22 estimated hours x 30	660
Bin Wang: labour	22 estimated hours x 30	660
Yonghui Dai: labour	22 estimated hours x 30	660
Tools	Python, Tkinter, SQLite, Git and existing editor/communication access; no extra purchase.	0
Resources	Assumed share of existing computer/network use.	30
Subtotal	1,980 + 0 + 30	2,010
Contingency	10% x 2,010	201
Current estimate	2,010 + 201	2,211

Change from the baseline

The original estimate was NZD 1,815. The extension adds NZD 360 labour and NZD 36 contingency, increasing the model by NZD 396. No additional server cost is included because the product remains local. The 12-hour allowance must be revised if actual work or review needs differ.

Limits and delivery risks

Risk	Response / coordinator
Unverified Windows setup	Yonghui runs the extracted ZIP on the presentation computer; the Windows launcher is reviewed, not execution-verified.
Existing data upgrade	Retain a database copy before important use; migration checks preserve original loan values and use blank legacy contacts.
Contact data used prematurely	Use fictional details; Xuerui seeks agreement on necessary collection and controls before real operation.
Client/adviser arrangement open	Confirm the assessed client role and appropriate Kaiārahi route; do not close review gates without participation.

Tax, depreciation, hosting and adviser fees are excluded. Agree any consultation arrangements and add actual new costs before contingency. Actual hours and expenditure remain to be recorded by the team.

11. Current acceptance testing record

Actual assistant execution: 24 October 2026 (Pacific/Auckland). Python 3.12.14, Darwin arm64, SQLite 3.53.1. Tests use disposable databases and fictional details. The source commit is recorded in section 9.

Check / story	Observed result	Status
AT01 / US01	Fresh database shows five available items and empty history.	PASS
AT02 / US02	Loan 1 created with borrower and phone; laptop marked On loan.	PASS
AT03 / US02	Duplicate borrow rejected without an additional record.	PASS
AT04 / US02	Unknown ID, blank, spaces-only, 81-character and control-character aliases rejected.	PASS
AT05 / US03	Return timestamp stored, laptop available and historical loan retained.	PASS
AT06 / US03	Repeated and nonexistent returns rejected; record count unchanged.	PASS
AT07 / US04	Empty, returned and active history states displayed correctly.	PASS
AT08 / US05	A separate process preserves loans, phone contacts and the five-item inventory.	PASS
AT09 / US06	Māia preserved in storage and UTF-8 CLI output after restart; physical display not assessed.	PASS
AT10 / UI	Non-numeric ID and invalid menu choice rejected; subsequent inventory action works.	PASS
AT11 / US01-US03	Tablet and Microphone can each be borrowed and returned using their loan IDs.	PASS
AT12 / US02/04/05	Missing/invalid phone contacts rejected; a formatted contact persists after restart.	PASS

Current result: all 12 technical acceptance checks passed. The unittest suite separately passed 12 business/storage checks and 8 GUI checks (20 total); exact methods are integrated in Appendix A.

Execution scope and acceptance limits

GUI interaction and small-window layout were exercised on macOS. Windows startup logic was reviewed but has not been executed on Windows. A copy of the existing local database was also upgraded and compared without changing original loan values; the live database was not overwritten. Client/community approval, real-data readiness and adviser-agreed acceptance remain open.

12. Evidence register and submission readiness

Item	Status	Supported conclusion
Current GUI/CLI and phone-enabled storage	ACTUAL reference artifact	Five devices, contacts, loan/return and retained history are implemented.
20 unit/GUI + AT01-AT12	ACTUAL assistant execution	Current observed behaviour on macOS, including phone checks and migration.
Release source d61c66ca11d2	ACTUAL Git object	13 product files match the source snapshot; historical commits are preserved.
Windows launcher	IMPLEMENTED; execution OPEN	Dependency selection and startup logic reviewed; a Windows machine check remains.
M01-M04, D01-D04, CF01-CF04	SIMULATED baseline case	A written example of requirements, reviews and scope decisions.
54 h baseline + 12 h extension	PLANNING ESTIMATE	66 h / NZD 2,211; no measured student time is claimed.
CG01-CG03 consultation	PREPARED; review OPEN	Questions and candidate gates; no adviser involvement is established.
Group/individual reflections	ROLE-BASED DRAFTS	Analysis for review, not personal statements supplied by the named students.

Submission package

Submit 01_Project_Report.pdf, 02_Product.zip and 03_Reflection_Report.pdf. The ZIP contains eight Python files, two platform launchers, README and two editable backlogs (13 files). presentation_script.md is a separate rehearsal aid. The GitHub root includes the source and these submission artifacts; runtime databases are excluded.

Participant and administrative details still needed

The team must carry out and evidence the required client-role/collaboration activities, follow the consultation route, confirm each member's contributions and adapt personal reflections to their own experience. Windows execution remains a technical follow-up. These are activities rather than missing prose.

[TO COMPLETE: student IDs, tutor and submission date]

Authorship and source

AI assistance produced the reference implementation, technical checks and document drafts. Actual assistant observations are labelled separately from the classroom scenario. Follow the course rules on assistance/disclosure. Assessment source: supplied Yoobee MSE800 Assessment 2 brief, pp. 1-7. The current owner requests, application, backlogs, executed checks and actual Git objects support the release description.

13. Task-by-task requirement check

The outline responds to all eight assessment tasks. Written coverage and technical checks do not substitute for required participant activities or guarantee a grade.

Task / weight	Current coverage	Evidence boundary
1. Team formation / 10%	Sections 1-2: roles, three-member arrangement and inclusive practices.	Roles are assigned; actual teamwork needs participant evidence.
2. Kickoff / 10%	Section 5: M01/D01-D04; section 8: consultation briefing.	Baseline meeting is simulated; actual consultation is open.
3. Requirements / 15%	Section 5: M02, questions, decisions and follow-up method.	Written scenario, not an actual participant meeting record.
4. Priorities / 10%	Sections 3/3.1: stories, current backlog, business value and impact assessment.	Current owner requests are recorded; client/community review remains open.
5. Agile / 15%	Sections 6-7: 14 baseline tasks/reviews; section 7.1: later work; section 9: Git.	The technical extension is actual assistant work, not an invented student sprint.
6. Cost / 10%	Section 10: 66-hour current estimate, labour/tools/resources and contingency.	NZD 2,211 is a planning estimate; actual time/cost is unconfirmed.
7. Testing / 10%	Sections 3/8/11 and Appendix A: criteria, 20 + 12 passing checks and open gates.	Windows execution, client/community acceptance and Kaiārahi criteria remain open.
8. Reflection / 20%	Separate group synthesis and three named role drafts cover four themes.	Members must replace proposed accounts with their actual experience.

Assessment and presentation scope

The required upload structure is two PDFs plus one runnable product ZIP. The brief also requires a final presentation of the product and development journey. The current presentation script demonstrates the GUI, contact field, returns and persistence, and explains the limits of technical evidence.

Tasks 2 and 7 request Kaiārahi consultation. The proposed internal client persona and prepared questions do not establish that participation. The Task 1 partnership and Task 4 local-priority conditions remain those in the brief.

Ethnic-diversity information is omitted from completion fields at the team's request. The original Task 1/rubric includes it; no tutor waiver is claimed. Te Tiriti and culturally informed teamwork remain part of the outline.

Appendix A1. Business and storage checks

Current assistant-run unittest output, 24 October 2026. The 12 methods below belong to test_system.LoanTests. The complete suite also includes the eight GUI methods on the next page.

Recorded method	Result
test_borrow_updates_availability_and_history	ok
test_double_borrow_is_rejected	ok
test_initial_inventory	ok
test_invalid_alias	ok
test_invalid_equipment_id	ok
test_legacy_database_is_upgraded_without_losing_loans	ok
test_missing_or_repeated_return	ok
test_new_equipment_can_be_borrowed_and_returned	ok
test_persistence_after_restart	ok
test_phone_validation_and_format_preservation	ok
test_return_keeps_history_and_allows_new_loan	ok
test_unicode_alias_round_trip	ok

Original runner summary for the current 20-check suite

```
Ran 20 tests in 0.836s
OK
```

Migration and historical evidence

The migration check constructs the earlier no-phone schema, opens it with the new repository and verifies unchanged loan IDs, names, borrowing/return timestamps and row counts on repeated openings. A separate copy of the existing database was compared before/after migration; original fields were preserved and absent contacts remained blank.

The 13 October baseline record contained 9 unit and 10 acceptance checks. It remains a historical observation of the earlier source; this report does not relabel those timings or results as the current 20 + 12 execution.

```
python3 -m unittest test_system -v
```

Appendix A2. GUI checks and repeatability

Current assistant-run methods in test_gui.GuiTests. They create real Tk widgets with disposable SQLite databases. All eight executed and passed on macOS; none was skipped in this recorded run.

Recorded method	Result
test_active_filter_and_restart_persistence	ok
test_alias_validation_and_missing_selection	ok
test_borrow_button_updates_tables_and_unicode	ok
test_inventory_and_empty_history	ok
test_phone_is_required_and_new_equipment_can_be_selected	ok
test_return_uses_loan_id_and_keeps_history	ok
test_small_window_keeps_history_and_actions_visible	ok
test_stale_inventory_rejects_duplicate	ok

What these checks establish

The GUI checks exercise selection, button invocation, saved names/phones, invalid input, duplicate protection after a stale selection, return by loan ID, active filtering, persistence and the smaller-window layout. The phone check also selects Tablet and Microphone. A successful result does not establish Windows behaviour or end-user approval.

Repeat the current suite

```
python3 -m unittest -v
python3 acceptance_checks.py
python3 acceptance_checks.py --output-dir checks
```

Run inside the extracted product directory. On Windows use py -3 in place of python3. The optional output directory receives a new JSON record and transcript with actual execution details and source hashes. GUI methods skip if Tk or a desktop display is unavailable; a skipped run is not GUI validation.

Appendix B. Start and retrieve the release

B1. Run the product ZIP

Extract 02_Product.zip completely into a writable folder. Open Campus_Equipment_Loan_System. Python 3.10+ with Tkinter and sqlite3 is required; no third-party Python packages or server are needed.

Platform	Start the desktop product
macOS	Double-click Start_GUI.command, or use bash Start_GUI.command. The launcher finds a suitable installed Python.
Windows	Double-click Start_GUI.bat. It checks py -3, python and python3. Read any dependency/error message before closing it. Windows execution remains to be verified.

```
python3 gui.py --db rehearsal.db
```

```
py -3 gui.py --db rehearsal.db
```

The first command is for a suitable Mac/Linux Python; the second is for Windows. Select equipment, enter a fictional alias and phone, and click Record loan. Select an active history row to return it. Restart with the same database to keep history. The default is equipment.db beside the source.

B2. Retrieve the exact product source

```
git clone --branch main https://github.com/dreamgithub0920/Campus_Equipment_Loan_System.git
```

```
cd Campus_Equipment_Loan_System
```

```
git checkout --detach d61c66ca11d2fd69c2bd40c005d3e4b53f65b100
```

```
python3 gui.py --db rehearsal.db
```

```
python3 -m unittest -v
```

```
python3 acceptance_checks.py
```

The cloned repository has the product files directly at its root: only one cd is needed. The pinned source commit includes the GUI and both launchers. Reports and the release ZIP are available on main in the later documentation commit. Public retrieval needs no sign-in.

B3. Existing data and terminal fallback

Keep an existing equipment.db to retain earlier records. Close the old app before reopening; migration adds the phone column and two devices without deleting existing loans. Historical blank contacts display Not recorded. Do not include a personal database in the submission ZIP.

```
python3 app.py --db rehearsal.db
```

The terminal fallback now asks for the phone after the alias. Use a separate demonstration database and fictional details. This local application has no shared server or automatic data synchronisation.